

1.5 HW. Write-up.

14. Use a truth table to prove the rule of deduction is valid. $p \leftrightarrow q, q \therefore p$

p	q	$p \leftrightarrow q$	$(p \rightarrow q) \wedge q$	$[(p \leftrightarrow q) \wedge q] \rightarrow p$
T	T	T	T	T
T	F	F	F	T
F	T	F	F	T
F	F	T	F	T

The rule is valid since the statement $[(p \leftrightarrow q) \wedge q] \rightarrow p$ is a tautology.

The conclusion, p , is true in every line that both of the premises are true.

40. Justify each step in the valid argument
 $(A \wedge B) \rightarrow C, B \vee D, \sim D, A \wedge E \therefore C$

1. $A \wedge E$ is a premise
2. A follows from ① by conjunctive simplification
3. $B \vee D$ is a premise
4. $\sim D$ is a premise
5. B follows from ③ and ④ by disjunctive syllogism
6. $A \wedge B$ follows from ② and ⑤ by conjunctive addition
7. $(A \wedge B) \rightarrow C$ is a premise
8. C follows from ⑥ and ⑦ by modus ponens